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SRI RAMACHANDRA FACULTY OF ENGINEERING AND TECHNOLOGY

ADA23DLU02 - Computer Vision Laboratory
PROJECT REVIEW

CrickSense - Ball-by-Ball Analysis of a Cricket Broadcast from a Single Camera Feed

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PROBLEM STATEMENT

- A cricket broadcast shows **one camera behind the bowler**, a score bar and sometimes a radar speed; where the ball pitched, its line and speed are measured only by **multi-camera installations** at top matches.
- The **ball is 6-15 pixels** across and moves 3-6 m between frames at 25 fps; the camera **pans and zooms** through every delivery, and the picture **cuts to replays** for most of each ball.
- Runs, wickets and extras follow the **Laws of Cricket** and often happen off screen, so they cannot be seen in the picture alone.
- Language models that write commentary **state facts that are not in their input**, so automated commentary cannot be trusted without checks.
- Therefore there is a need for a system that measures each ball from the **broadcast alone**, keeps a **checked match state**, and speaks **fact-checked commentary** - offline and live.

INTRODUCTION

CrickSense is a computer-vision system that analyses a cricket match from its **television broadcast alone** - one camera feed, no sensors and no access to the broadcaster's data. It calibrates the pitch automatically from its **painted creases**, reads the broadcaster's **score bar**, detects and tracks the players and names them by role, and finds each delivery's **bounce, length, line and speed**.

A score engine keeps the match state by the **Laws of Cricket** and checks it against the score bar after every ball. From these measured facts, a language model writes ball-by-ball commentary in which **every line is fact-checked**, spoken by a local text-to-speech model over the crowd. A **live mode** runs the time-critical parts as the video plays, on one laptop GPU.

OBJECTIVES

- To **calibrate the broadcast camera** to the pitch automatically, every frame, without hand-clicked points.
- To **read the score bar** reliably over a whole match and use it as the source of outcomes.
- To **detect and track the players** and give them roles and names.
- To find each delivery's **bounce, length, line and speed**, and measure their accuracy against radar.
- To keep **score, strike and bowler figures by the Laws**, checked against the bar after every ball.
- To generate **fact-checked commentary and speech** and render the annotated video.
- To run outcomes, speed and named players **live**, and measure the latency.

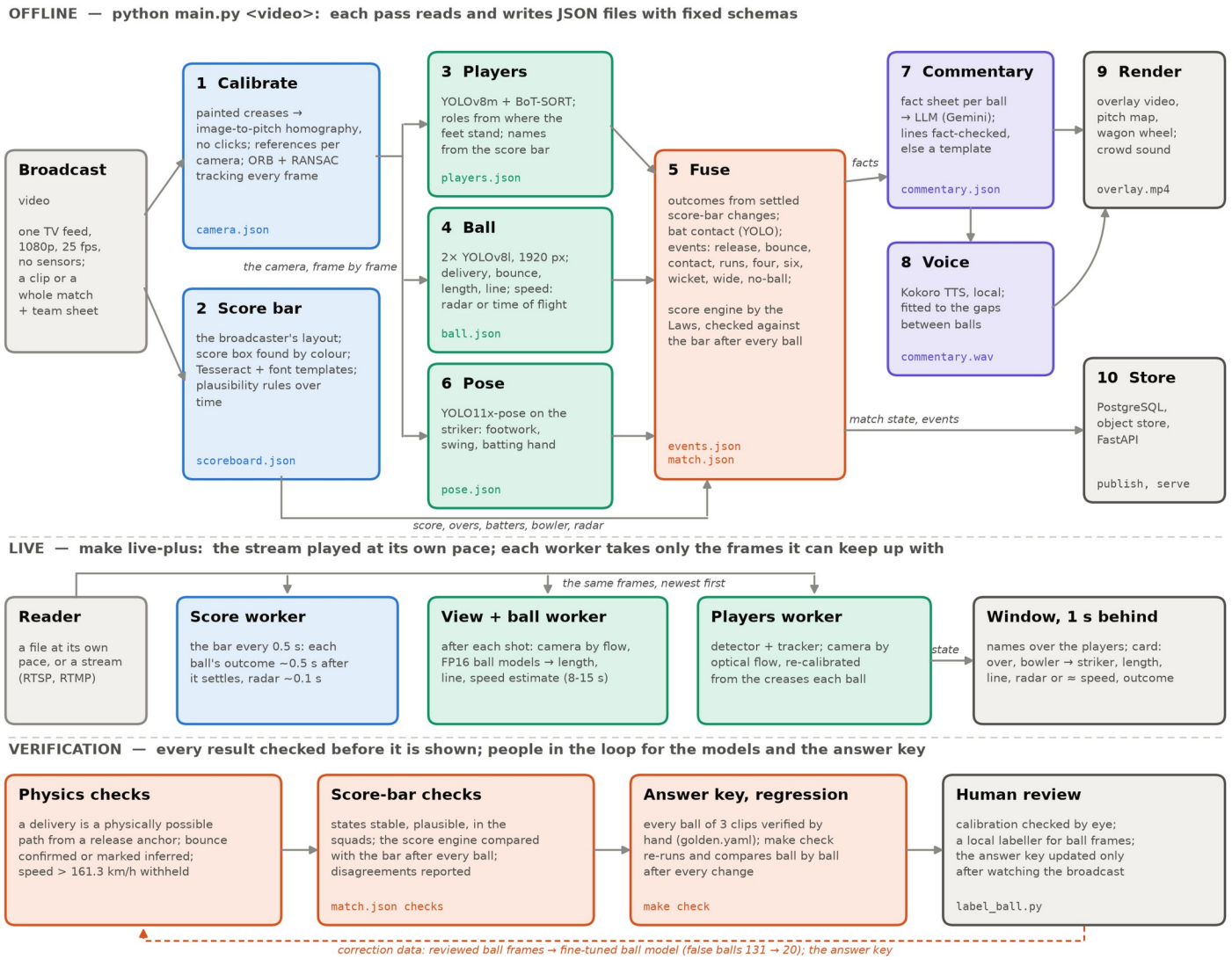
BASIC RESEARCH SURVEY

S.No	Paper Title & Year	Author & Publication	Technique / Methodology	Advantages	Limitations
1	Deep-learning-based segmentation of ball deliveries and tracking in cricket (2022)	K. Abbas et al., arXiv	YOLO and MobileNet for delivery segmentation; RetinaNet for ball localisation	Robust delivery segmentation; curated cricket dataset	No pitch calibration, so no length, line or speed
2	Hawk-Eye tennis system (2003)	N. Owens, C. Harris, C. Stennett, IEE VIE	Calibrated high-speed cameras; 3D ballistic reconstruction	Broadcast-grade bounce and trajectory accuracy	Needs an installation at the venue; expensive
3	TrackNet: tracking high-speed and tiny objects in sports (2019)	Y.-C. Huang et al., IEEE AVSS	Heatmap CNN over consecutive frames	Finds tiny, blurred balls; handles short occlusions	Fixed camera; needs labelled sequences
4	TTNet: real-time temporal and spatial video analysis of table tennis (2020)	R. Voeikov et al., IEEE/CVF CVPR Workshops	Multi-task network: ball detection, event spotting, segmentation	Real-time; detects events as well as the ball	Fixed camera; another sport

BASIC RESEARCH SURVEY

S.No	Paper Title & Year	Author & Publication	Technique / Methodology	Advantages	Limitations
5	Sports field localization via deep structured models (2017)	N. Homayounfar et al., IEEE CVPR	Field-line segmentation with a structured model of camera parameters	Calibrates broadcast frames with no clicks	Needs a large field with many lines in view
6	Sports camera calibration via synthetic data (2019)	J. Chen, J. J. Little, IEEE/CVF CVPR Workshops	Learned from synthetic views of the field; feature retrieval and refinement	Works on broadcast football	Needs a field model and training data
7	BoT-SORT: robust associations multi-pedestrian tracking (2022)	N. Aharon et al., arXiv	Kalman tracking with camera-motion compensation and re-identification	Strong multi-object tracking accuracy	Camera motion from image features; no roles
8	CricShotClassify: classifying batting shots from cricket videos (2021)	A. Sen et al., Sensors	CNN with a gated recurrent unit on shot clips	Good accuracy on labelled shot clips	Needs a labelled shot set; no pitch geometry

SYSTEM ARCHITECTURE



PROPOSED METHODOLOGY

Video → **Calibrate** → **Score bar** → Players → Ball → Pose → Fuse → Commentary → Voice → Render → Live

- **Input:** a broadcast clip or a whole match (MP4, 1080p, 25 fps) or a live stream, with the match's team sheet.
- **Calibration:** white-line candidates from the **painted creases** give hypotheses for the crease corners; each defines a **homography** from image pixels to pitch metres, scored by how much of the regulation creases lands on paint. A camera view is accepted only with two strong agreeing frames.
- **Camera tracking:** every frame is registered to the reference views with **ORB features and RANSAC**, and smoothed with a Savitzky-Golay filter.
- **Score bar:** the broadcaster's layout fields and the score box found by colour, read with **Tesseract and font templates**; a state is trusted only if stable and plausible.

TOOLS AND TECHNIQUES

- **Tools**

Python, OpenCV, PyTorch, Ultralytics YOLOv8 / YOLO11, BoT-SORT, Tesseract OCR, Gemini API, Kokoro TTS, Demucs, FFmpeg, PostgreSQL, FastAPI

- **Techniques**

Homography calibration from the painted creases, ORB + RANSAC camera tracking, optical flow, multi-object tracking with role assignment, physically constrained ball tracking and time-of-flight speed, OCR with font templates, a rule-based score engine, fact-checked language generation, FP16 inference

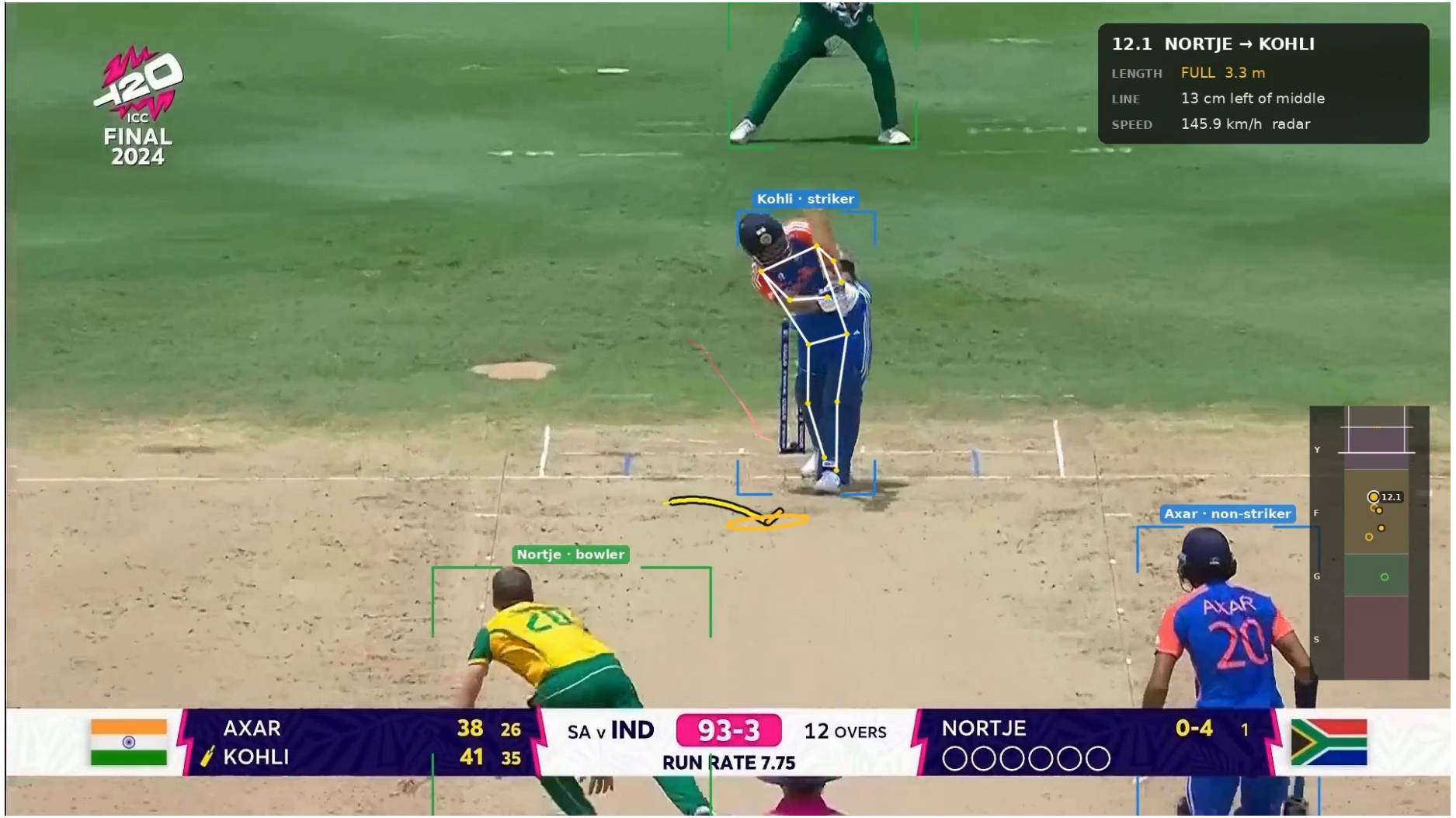
- **Datasets**

ICC Men's T20 World Cup 2024 broadcasts (the final and a Super 8 match); three hand-verified answer-key clips; 20 random benchmark clips; 94 hand-reviewed frames for fine-tuning the ball model

RESULTS

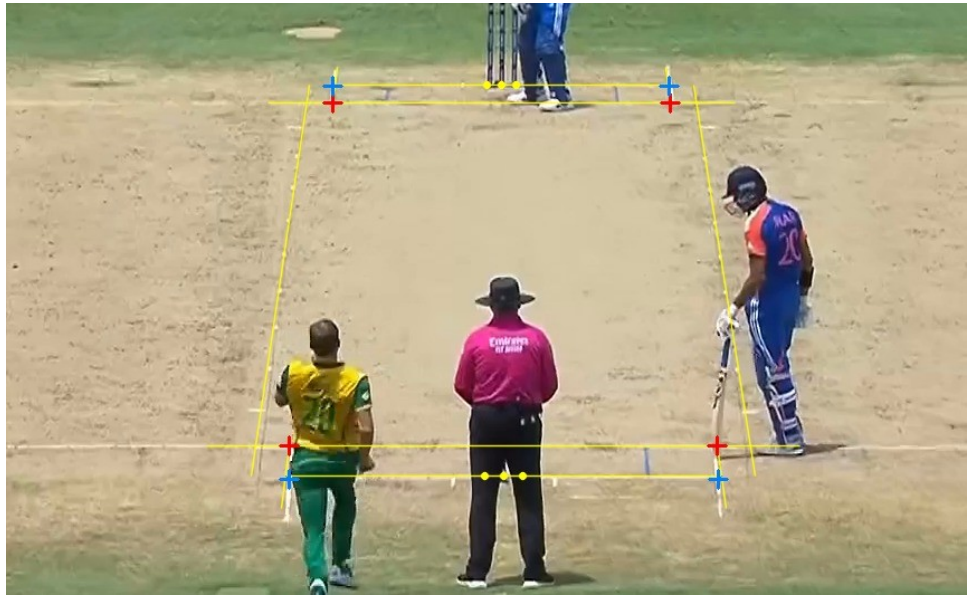
Measure	Result
Calibration vs hand-clicked crease corners	7 of 8 corners within 3.3 px
20 random clips of the final, no clicks	14 calibrated, none wrongly
Score bar over two whole matches	all 4 final scores exact (176-7, 169-8, 205-5, 181-7)
Deliveries on three answer-key clips	21 / 21 found, 21 / 21 outcomes right
Score engine against the score bar	230 / 230 checks agree
Our speed estimate vs radar	5.7% median error on the final (8.3% over both matches)
Player identities (development clip)	51 identities for 51 appearances
TV commentators removed (Demucs, 2 passes)	voice between our lines: 213 s → 0 s

RESULTS

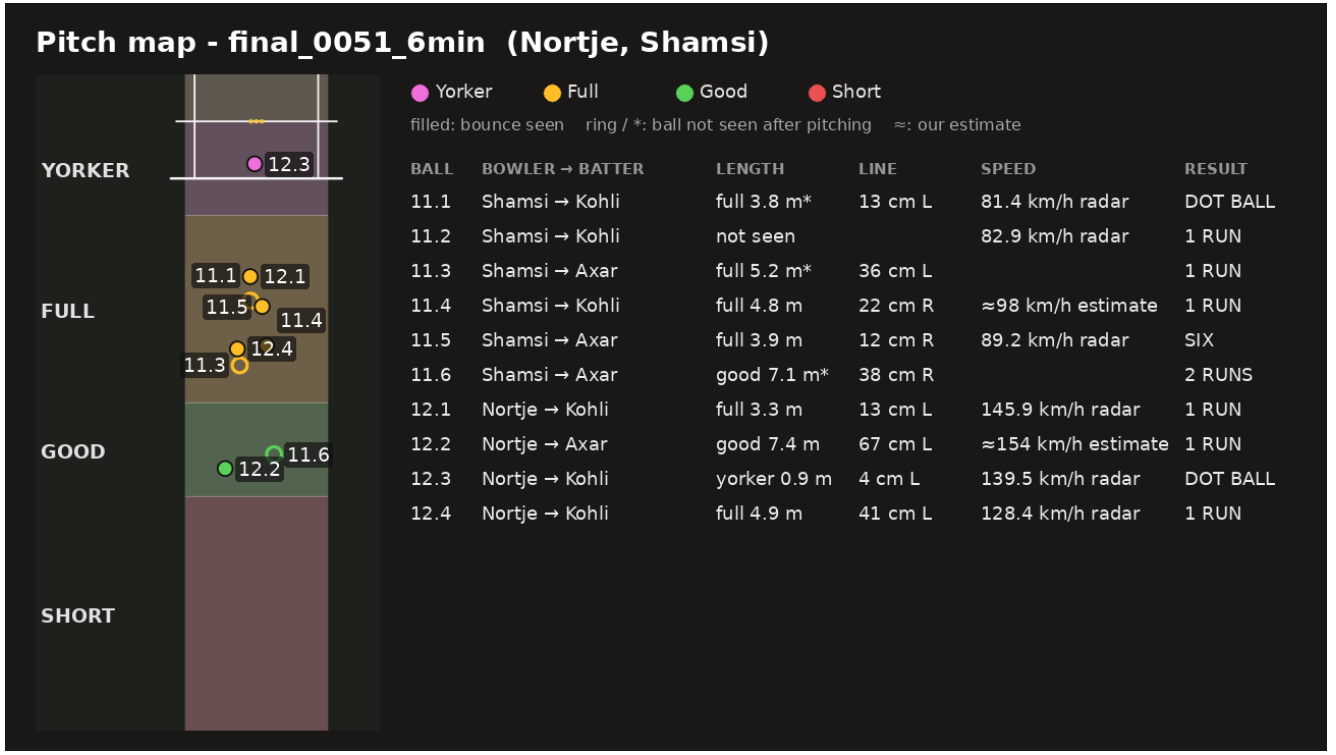


Ball 12.1, Nortje to Kohli: named players, the striker's pose, the ball's path and bounce, and the delivery card

RESULTS



Automatic calibration: creases projected through the homography



Pitch map: bounce, length, line, speed and result per ball



Score-bar fields read every 0.5 s

RESULTS

outputs/st_bumrah_o2/ball.json - delivery 1.1


```json
{
  "delivery_id": 0,
  "release_frame": 298,
  "release_t": 11.9,
  "release_point": [18.66, -0.8, 2.0],
  "detections": 12,
  "bounce_t": 12.2,
  "bounce_xy": [7.4, -0.08],
  "bounce_confirmed": true,
  "length_m": 7.4,
  "line_m": -0.08,
  "length_class": "good",
  "radar_speed_kph": 139.3,
  "speed_estimate_kph": 143.88,
  "track_model": "ball_ft.pt"
}
```


ball.json: a delivery's geometry and speed

- Every pass writes **JSON with a fixed schema**; the next pass reads it.
- **ball.json:** release, bounce, **length 7.40 m**, line, radar **139.3** vs our **143.9 km/h**.
- **events.json:** each ball's events, with their source and confidence.

outputs/st_bumrah_o2/events.json - the four (1.4) and the wicket (1.3)


```json
[
  {
    "frame_id": 5760,
    "t": 230.46,
    "type": "RUNS",
    "payload": {
      "delivery_id": 3,
      "bowler": "BUMRAH",
      "striker": "MARKRAM",
      "over": "1.4",
      "runs": 4,
      "off_bat": 4,
      "extras": 0,
      "kind": "legal",
      "score": "11-1"
    },
    "confidence": 0.9,
    "source": "scoreboard"
  },
  {
    "frame_id": 2412,
    "t": 96.5,
    "type": "WICKET",
    "payload": {
      "delivery_id": 2,
      "bowler": "BUMRAH",
      "striker": "HENDRICKS",
      "over": "1.3",
      "score": "7-1"
    },
    "confidence": 0.9,
    "source": "scoreboard"
  }
]
```


events.json: the four and the wicket

RESULTS

outputs/st_bumrah_o2/match.json - the state after the wicket

```json
{
  "over": "1.3",
  "runs": 7,
  "wickets": 1,
  "legal_balls": 9,
  "striker": "MARKRAM",
  "non_striker": "DE KOCK",
  "bowler": "BUMRAH",
  "batters": {
    "HENDRICKS": [4, 5],
    "DE KOCK": [1, 4],
    "MARKRAM": [0, 0]
  },
  "bowlers": {
    "BUMRAH": [0, 3, 1]
  },
  "source": "engine",
  "checks": {
    "same bowler within the over": true,
    "striker as predicted": true,
    "score": true,
    "wickets": true,
    "overs": true,
    "Markram runs/balls": true,
    "De Kock runs/balls": true,
    "on strike": true,
    "Bumrah figures": true
  },
  "notes": ["wicket: incoming batter and strike from the score bar"]
}
```

match.json: the state after the wicket, every check true

outputs/replay-super-8s_i1_o3/commentary.json - Starc to Rohit, 2.1

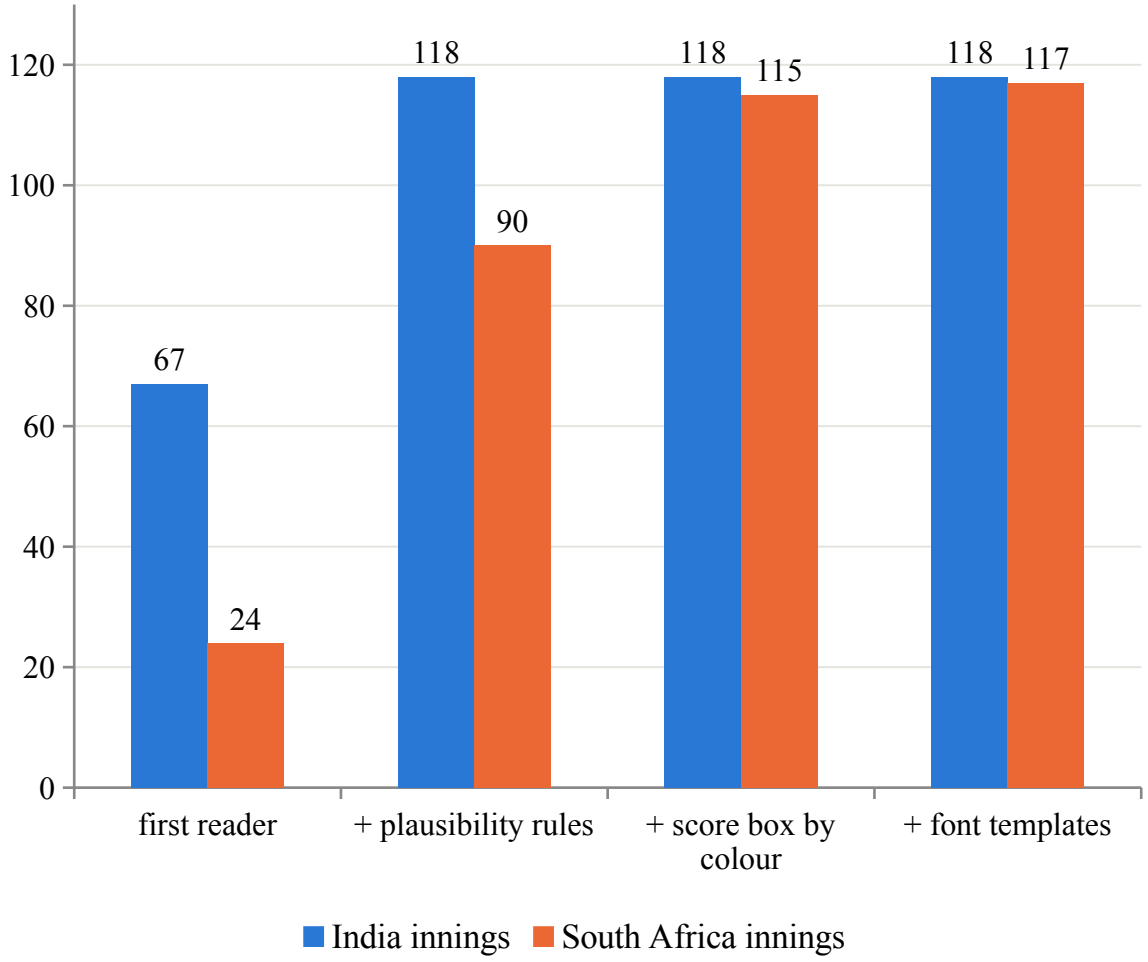
```json
{
  "engine": "gemini:gemini-3.5-flash-lite",
  "facts": {
    "over": "2.1",
    "bowler": "Starc",
    "striker": "Rohit",
    "length_class": "full",
    "speed_kph": 138.3,
    "speed_source": "radar",
    "runs": 6,
    "boundary": "SIX",
    "score": "12-1"
  },
  "lines": [
    {
      "kind": "ball",
      "source": "llm",
      "text": "Starc bowls a full delivery at 138.3 kph, Rohit plays it from the crease for SIX!"
    },
    {
      "kind": "after",
      "source": "llm",
      "text": "India move to 12-1."
    }
  ]
}
```

commentary.json: the facts and the line written from them

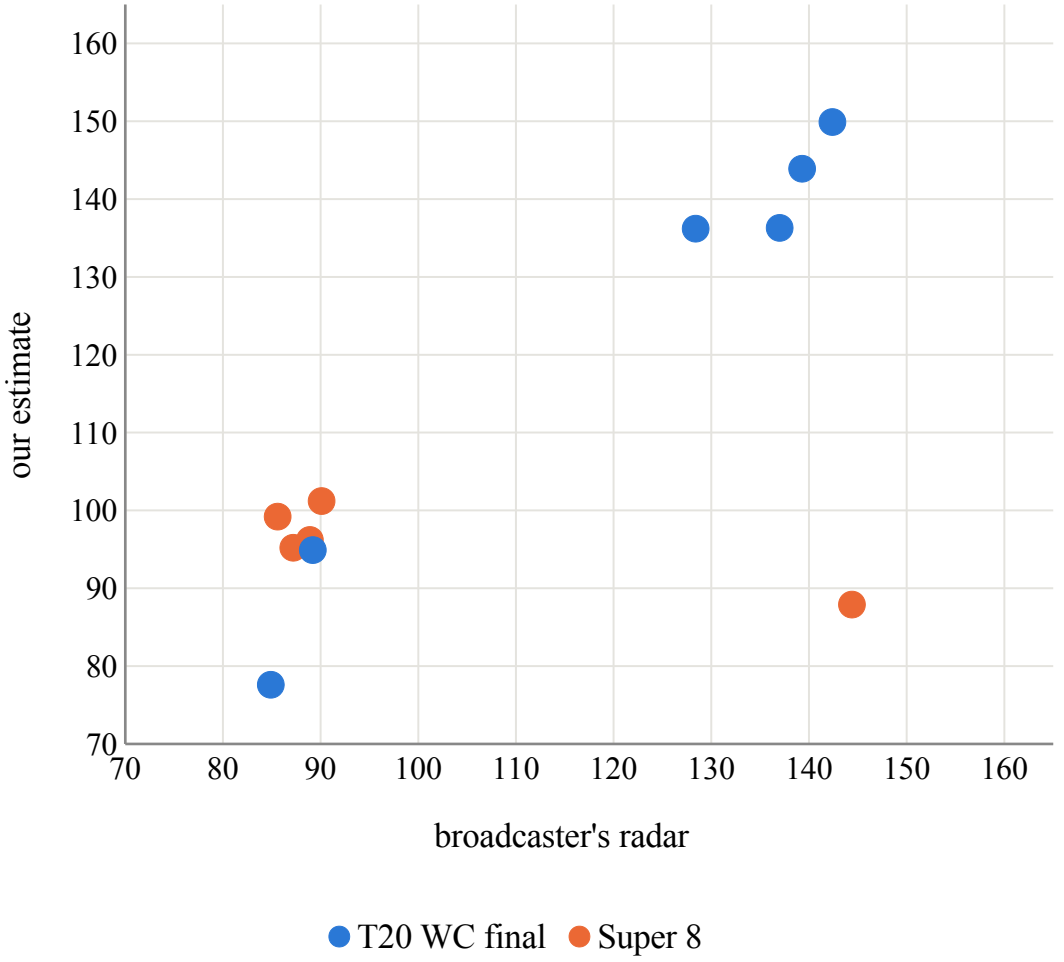
- **match.json** is checked against the broadcaster's score bar after every ball - here all true.
- **commentary.json:** the line passed the **fact check** - every number and name is in the facts.

RESULTS

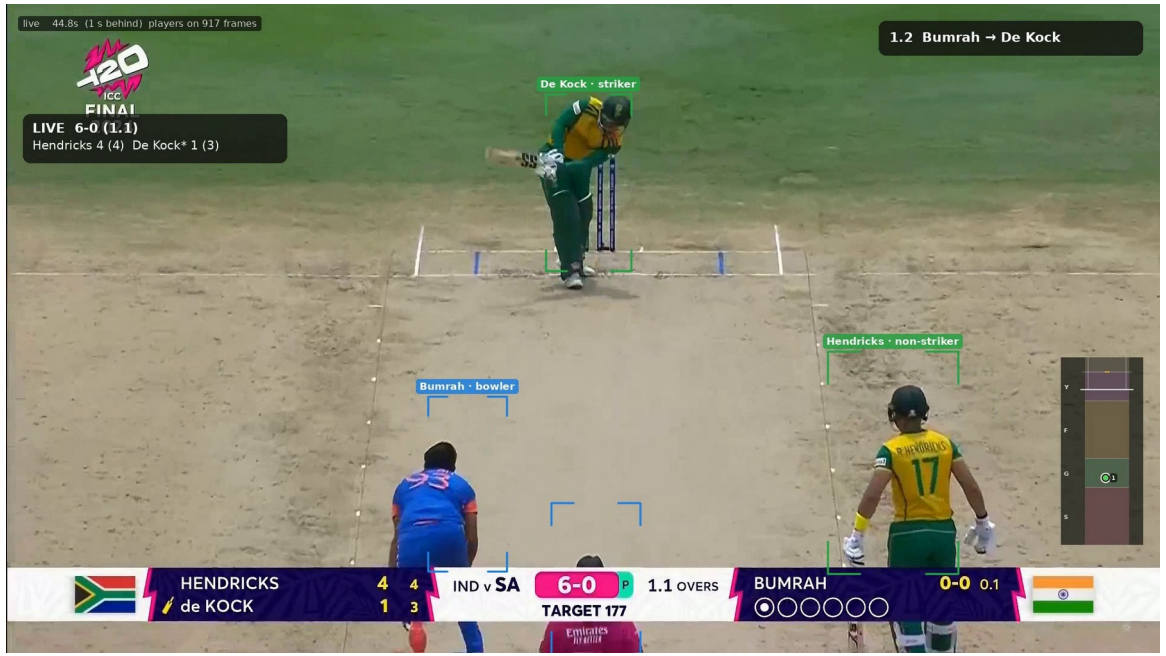
Score-bar ball counts seen (of 121), whole final



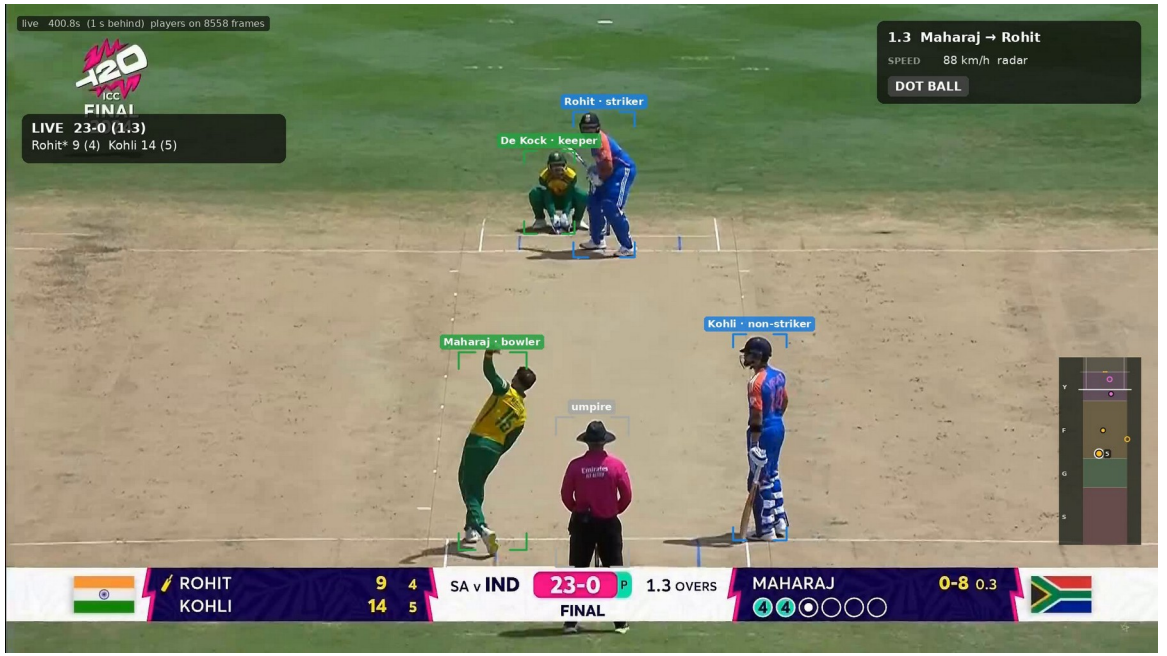
Our time-of-flight speed vs radar (km/h)



RESULTS



Live, Bumrah's over: striker, non-striker, bowler and umpire named



Live, the whole final: names re-calibrated from the creases

- Live on one laptop GPU: each ball's **outcome 0.5 s** after the score bar, **radar 0.1 s**, length, line and our speed **12-23 s** after the ball.
- **FP16** made the ball model 3.3× faster; the **optical-flow camera** stays within 0.25 m of the offline calibration.

FUTURE SCOPE

- Use **50 fps footage** and labelled frames of the ball's first flight to improve pace-bowling speed and recover missed deliveries.
- Make crease calibration faster so that players are **named live through a whole match**.
- Build a **labelled answer key for shot types**, so that a vision-language model can be scored before it is used.
- Add **score-bar layouts for other broadcasters** and a short ball-labelling round for each new broadcast.
- Use **multi-camera input** where available for 3D ball trajectories.

CONCLUSION

In conclusion, **CrickSense** measures a cricket match from its broadcast alone. It calibrates the pitch from its painted creases, reads the score bar over whole matches, names the players by role, and finds each delivery's length, line and speed, keeping a match state by the Laws of Cricket.

On held-out clips all **21 deliveries and outcomes** were right, the score engine agreed with the bar on **230 of 230 checks**, both matches' final scores were read exactly, and our speed estimate was within **5.7% of radar** on the final. With fact-checked commentary and a live mode on one laptop GPU, CrickSense turns a broadcast into verified ball-by-ball analysis.

TIMELINE

27/09/26 - 29/09/26 (3 days)	Package skeleton, calibration and camera tracking; players pass with roles and names; ball pass
30/09/26 - 01/10/26 (2 days)	Fuse and score engine, overlay, commentary and voice; automatic calibration; storage, API and live mode
02/10/26 - 14/10/26 (13 days)	Pace (Bumrah's over), multi-reference calibration, score-bar font templates, answer-key regression check
15/10/26 - 20/10/26 (6 days)	Live mode re-measured, voice options, the second match (Super 8), the one-command pipeline
21/10/26 - 28/10/26 (8 days)	Commentary on Gemini, TV commentators removed with Demucs, live-plus with named players
29/10/26 (1 day)	Live naming over a whole match (crease re-calibration); project report

GUIDE DETAILS

Guide

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THANK YOU